275-276页7,8,10,12; 289页5.
276页9; 289-290页4,11,14.

275-7

Let $\langle\cdot,\cdot\rangle$ be the inner product on $\mathbb{C}^2$, prove that there is no non-zero linear operator on $\mathbb{C}^2$ such that $\langle\alpha, T\alpha\rangle = 0$ for every $\alpha \in \mathbb{C}^2$. Generalize.
Proof: Take a basis e_1, e_2 such that T is upper-triangular, then $\langle e_1, Te_1\rangle = 0$ implies $Te_1 = 0$, and likewise $\langle e_2, Te_2\rangle = 0$ implies $Te_2 = 0$, hence $T = 0$. This holds for general $T \in \mathcal{L}(\mathbb{C}^n)$.

276-8

Let $A \in \mathbb{R}^{2\times 2}$, for $X, Y \in \mathbb{R}^{2\times 1}$, let $f_A(X, Y) = Y^t AX$. Show that f_A is an inner produce iff $A = A^t$, $A_{11}, A_{22} > 0$ and $\det A > 0$.
Proof: $A = A^t$ iff $f_A(X, Y) = f_A(Y, X)$, $A_{11}, A_{22} > 0, \det A > 0$ iff $f_A(X, X) \geqslant 0$ and $f_A(X, X) = 0 \iff X = 0$.

276-12

Let V be finite dim and $\mathcal{B} = \{\alpha_1, \cdots, \alpha_n\}$ be a basis of V. Let $\langle\cdot,\cdot\rangle$ be an inner product for V. If $c_1, \cdots, c_n$ are any n scalars, show that there is exactly one vector $\alpha \in V$ such that $\langle\alpha, \alpha_j\rangle = c_j$.
Proof: Suppose $a_{ij} = \langle\alpha_i, \alpha_j\rangle$, then for $\alpha = \sum x_i\alpha_i$, $\langle\alpha, \alpha_j\rangle = \sum_i x_i a_{ij}$. Hence α corresponds to the solution to $XA = Y$ where $A = (a_{ij})$ and $C = (c_j)$, so there is a unique α.

289-5

Let V be an inner product space, $\alpha, \beta \in V$. Show that $\alpha = \beta$ iff $\langle\alpha, \gamma\rangle = \langle\beta, \gamma\rangle$ for every $\gamma \in V$.
Proof: If $\langle\alpha, \gamma\rangle = \langle\beta, \gamma\rangle$, let $\gamma = \alpha - \beta$, then $\langle\gamma, \gamma\rangle = \langle\alpha, \gamma\rangle - \langle\beta, \gamma\rangle = 0$ so $\gamma = 0$ and $\alpha = \beta$.
The converse is trivial.

276-9

Show that the quadratic form determined by the inner product satisfies the parallelogram law:

$$|\alpha + \beta|^2 + |\alpha - \beta|^2 = 2|\alpha|^2 + 2|\beta|^2.$$

Proof: $\langle\alpha + \beta, \alpha + \beta\rangle = \langle\alpha, \alpha\rangle + \langle\alpha, \beta\rangle + \langle\beta, \alpha\rangle + \langle\beta, \beta\rangle$ and $\langle\alpha - \beta, \alpha - \beta\rangle = \langle\alpha, \alpha\rangle + \langle\beta, \beta\rangle - \langle\alpha, \beta\rangle - \langle\beta, \alpha\rangle$.
Hence $|\alpha + \beta|^2 + |\alpha - \beta|^2 = 2|\alpha|^2 + 2|\beta|^2$.

289-11

Let $\{\alpha_1, \cdots, \alpha_n\}$ be an orthonormal basis for V, show that for any $\alpha, \beta \in V$,

$$\langle\alpha, \beta\rangle = \sum_{k=1}^{n} \langle\alpha, \alpha_k\rangle\overline{\langle\beta, \alpha_k\rangle}.$$

Proof: $\alpha = \sum \langle\alpha, \alpha_k\rangle\alpha_k$ and $\beta = \sum \langle\beta, \alpha_k\rangle\alpha_k$ so

$$\langle\alpha, \beta\rangle = \sum_{j,k} \langle\alpha, \alpha_j\rangle \cdot \overline{\langle\beta, \alpha_k\rangle} \cdot \langle\alpha_j, \alpha_k\rangle = \sum_{k=1}^{n} \langle\alpha, \alpha_k\rangle\overline{\langle\beta, \alpha_k\rangle}.$$

289-14

Following Ex11, let $T \in \mathcal{L}(V)$ and $A = T_{\mathcal{B}}$. Prove that $A_{ij} = \langle T\alpha_j, \alpha_i\rangle$.
Proof: $T\alpha_j = \sum A_{ij}\alpha_i$ so $A_{ij} = \langle T\alpha_j, \alpha_i\rangle$.